

SUPRATIK BHOWAL

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SKILLS

Programming Languages: Python, Objective-C, MATLAB, Java, HTML5

ML/DL: PyTorch, TensorFlow, SciPy, OpenCV, Scikit-Learn, HuggingFace Transformers, Kaggle, Weights & Biases, JAX,

Technologies & Frameworks: LangChain, LangGraph, FastAPI, Git/GitHub, Computer Vision, RL, SciML, MySQL, GCP

Languages: English (Full Professional Proficiency), Bengali (Native), Hindi (Professional Working Proficiency)

EDUCATION

Institute of Engineering and Management

B.Tech Computer Science (AI & ML): YGPA – 9.11 (1st yr)

Kolkata, India

2024 - Present

South Point High School

PCM with Computer Science: %age – 90% (12th) & 95 % (10th)

Kolkata, India

2010 - 2024

EXPERIENCE

Rouge International & Rouge Ventures — Associate Intern

Remote

Developed and deployed AI tools for a VC firm and led team training on applied AI workflows.

Apr 2026 – Present

Asvix — AI Intern

Kolkata, India

Engineered a domain-specific chatbot for IGNOU using RAG, PDF parsing, and fine-tuned LLM responses

Apr 2026 – Present

Media Nirvana — AI Intern

Kolkata, India

Built end-to-end AI infrastructure for startups including CRM & WhatsApp bots, lead capture systems, a redesigned website and SEO/AEO/GEO-optimized content pipelines.

Jan 2026 – Present

IEM with IIT Mandi Chapter — HCI Research Intern

Kolkata, India

Conducted research in Human-Computer Interaction, focusing on innovative interfaces and robot training.

July 2025 – June 2026

IEM Centre of Excellence for Data Science (CEDS) Winter Internship — Research Intern

Kolkata, India

Created a Project on 'Forward Collision Avoidance in Autonomous Driving' using ADAS & computer vision.

Jan 2026 – Feb 2026

IEEE Engineering Medicine and Biology Society (EMBS) — Student Research Intern

Pune, India

Developed DL algos for 'Epileptic Seizure Detection and Prediction' under mentorship of Prof. Sneha Deshmukh.

Jun 2025 – Jul 2025

PROJECTS

Agentic HR Operations Assistant: Python, NLP, RAG, Faiss Vector Database, PyTorch

NLP-powered system that answers complex HR policy queries using RAG and executes HR actions from natural language.

HarmonyRL: Python, HuggingFace, PyTorch, Scipy, Librosa, RL

Deep Learning framework for generating symbolic music (MIDI) using Supervised Learning, Reinforcement Learning (RL), and Diffusion-based Postprocessing.

Epilepsy Prediction & Detection: Python, Scikit-learn, TensorFlow, Scipy

DL algorithms for real-time Seizure Detection/Prediction on EEG signals. CNN-BiLSTM achieved 98% prediction accuracy, while the Transformer model showed 89% detection accuracy. (IEEE EMBS Internship)

Brain MRI Tumor Classification: Python, PyTorch, OpenCV, Scikit-learn

HOG + LBP feature fusion with tuned SVM (RBF kernel) for multi-class brain tumor classification from MRI scans.

BrandSCOPE: Python, LangChain, JavaScript, BeautifulSoup, OpenAI API

Self-hosted brand content engine that scrapes brand DNA, researches live trends, and writes SEO/AEO/GEO-optimized articles in the brand's own voice.

Sentiment Analysis & Recommendation: Python, Scikit-learn, Tensorflow

Sentiment Classifier to analyze and predict the sentiment of user reviews collected from 1 million Google Play Store entries.

CERTIFICATIONS

- Build SLM from Scratch – Vizura Technologies
- Hand-on approach to AI for real-world application (AI4ICPS Cohort 2) – IIT Kharagpur AI4ICPS
- Introduction to Natural Language Processing – NPTEL
- Neural Networks for Computer Vision and Natural Language Processing – NPTEL
- Cyber Security Fundamentals – University of London